12. What to Buy? A marketing research experiment was conducted to study the relationship between the length of time necessary for a buyer to reach a decision and the number of alternative package designs of a product presented. The products were identical except for the package design. The length of time necessary to reach a decision was recorded for 15 participants in the marketing research study.

DATA SET DS1219

Length of Decision Time, y (sec)	5, 8, 8, 7, 9	7, 9, 8, 9, 10	10, 11, 10, 12, 9
Number of Alternatives, x	2	3	4

a. Find the least-squares line appropriate for these data.
b. Plot the points and graph the line as a check on your calculations.
c. Calculate s^2.
d. Do the data present sufficient evidence to indicate that the length of decision time is linearly related to the number of alternative package designs? (Test at the $\alpha = .05$ level of significance.)
e. Find the approximate p-value for the test and interpret its value.
f. If they are available, examine the diagnostic plots to check the validity of the regression assumptions.
g. Estimate the average length of time necessary to reach a decision when three alternatives are presented, using a 95% confidence interval.

a) $\Sigma x = (5\times2)+(5\times3)+(5\times4)$
$= 45$
$\bar{x} = \frac{45}{3} = 15$
$\Sigma x^2 = (5\times2^2)+(5\times3^2)+(5\times4^2)$
$= 145$
$\Sigma y = (5+8+8+7+9)+(7+9+8+9+10)+(10+11+10+12+9)$
$= 132$
$\Sigma y^2 = 1204$
$\Sigma xy = 2(37)+3(43)+4(52)$
$= 411$
$S_{xx} = 145 - \frac{45^2}{15}$
$= 10$
$S_{yy} = 1204 - \frac{132^2}{15}$
$= 42.4$
$S_{xy} = 411 - \frac{45\times132}{15}$
$= 15$
$b = \frac{15}{10} = 1.5$
$a = 8.8 - (1.5\times3)$
$= 4.3$
$\hat{y} = 4.3 + 1.5x$

c) $SSE = S_{yy} - b\,S_{xy}$
$= 42.4 - (1.5\times15)$
$= 19.9$
$s^2 = \frac{19.9}{15-2}$
$= 1.5308$

d) $H_0: b = 0$
$H_1: b \neq 0$

$t = \frac{b}{se(b)} = \frac{b}{\sqrt{\frac{s^2}{S_{xx}}}} = \frac{1.5}{\sqrt{\frac{1.5308}{10}}}$
$= \frac{1.5}{0.3913} = 3.833$

$df = n-2 = 13$
$t_{0.025;13} = 2.160$

$\therefore$ since $|t| = 3.833 > 2.160$, reject H_0

DATA SET DS1221

15. Strawberries III The following data (Exercise 16, Section 12.2) were obtained in an experiment relating the dependent variable y (texture of strawberries) with x (coded storage temperature).

x	−2	−2	0	2	2
y	4.0	3.5	2.0	0.5	0.0

a. Estimate the expected strawberry texture for a coded storage temperature of $x = -1$. Use a 99% confidence interval.

b. Predict the particular value of y when $x = 1$ with a 99% prediction interval.

c. At what value of x will the width of the prediction interval for a particular value of y be a minimum, assuming n remains fixed?

a) $\sum x = -2-2+0+2+2$

$= 0$

$\bar{x} = 0$

$\sum y = 4 + 3.5 + 2.0 + 0.5 + 0$

$= 10$

$\bar{y} = \frac{10}{5} = 2$

$\sum y^2 = 32.5$

$\sum xy = -14$

$S_{xx} = 16 - 0 = 16$

$S_{yy} = 32.5 - \frac{10^2}{5} = 12.5$

$S_{xy} = -14 - 0 = -14$

$b = \frac{-14}{16} = -0.875$

$a = 2 - (-0.875 \times 0)$

$= 2$

$\hat{y} = 2 - 0.875x$

when $x = -1$,

$y = 2 - 0.875(-1)$

$= 2.875$

when $\alpha = 0.01$,

$t_{0.005,3} = 5.841$

$SSE = S_{yy} - bS_{xy}$

$= 12.5 - (-0.875 \times -14)$

$= 0.25$

$s^2 = \frac{SSE}{n-2} = \frac{0.25}{5-2} = 0.0833$

$s = \sqrt{0.0833} = 0.2887$

$CI = 2.875 \pm 5.841 \times 0.2887 \times \sqrt{\frac{1}{5} + \frac{(-1-0)^2}{16}}$

$= 2.875 \pm 0.864$

b) when $x = 1$,

$y = 2 - 0.875(1)$

$= 1.125$

$PI = 1.125 \pm 5.841 \times 0.2887 \times \sqrt{1 + \frac{1}{5} + \frac{(1-0)^2}{16}}$

$= 1.125 \pm 1.895$

$PI = (-0.770, 3.020)$

c) width of PI $= \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}$

since $1, n, S_{xx}$ are fixed, it reaches min when $(x_p - \bar{x})^2$ reaches min.

$\therefore$ min is $x_p - \bar{x} = 0$, $x_p = \bar{x}$

DATA SET DS1223

18. Plant Science An experiment was conducted to determine the effect of various levels of phosphorus on the inorganic phosphorus levels in Sudan grass. The data in the table represent the levels of inorganic phosphorus in micromoles (μmol) per gram dry weight. Use the *MINITAB* output to answer the questions.

Phosphorus Applied, x	Phosphorus in Plant, y
.50 μmol	204
	195
	247
	245
.25 μmol	159
	127
	95
	144
.10 μmol	128
	192
	84
	71

a. Plot the data. Do the data appear to exhibit a linear relationship?

b. Find the least-squares line relating the plant phosphorus levels y to the amount of phosphorus applied to the soil x. Graph the least-squares line as a check on your answer.

c. Do the data provide sufficient evidence to indicate that the amount of phosphorus present in the plant is linearly related to the amount of phosphorus applied to the soil?

d. Estimate the mean amount of phosphorus in the plant if .20 μmol of phosphorus is applied to the soil. Use a 90% confidence interval.

Regression Analysis: y versus x

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	80.85	22.40	3.61	0.005
x	270.82	68.31	3.96	0.003

Regression Equation

y = 80.9 + 270.8 x

Settings

Variable	Setting
x	0.2

Prediction

Fit	SE Fit	90% CI	90% PI
135.015	12.6264	(112.130, 157.900)	(60.6448, 209.386)

$\Sigma x = 3.40$

$\bar{x} = 0.28333$

$\Sigma x^2 = 1.29$

$\Sigma y = 1891$

$\bar{y} = 157.58333$

$\Sigma y^2 = 357791$

$\Sigma xy = 624.25$

$S_{xx} = 0.32667$

$S_{yy} = 59800.9167$

$S_{xy} = 88.4667$

b. $b = \frac{S_{xy}}{S_{xx}}$

$= \frac{88.4667}{0.32667} = 270.81$

$a = 157.58333 - (270.81 \times 0.28333) = 80.85$

$\hat{y} = 80.853 + 270.814x$

c. $SSE = 59800.9167 - (270.814)(88.4667)$

$= 59800.9$

$s^2 = \frac{35842.8836}{10}$

$= 3584.2864$

$s = 59.86$

$t = \frac{270.8143}{\sqrt{\frac{3584.2864}{0.32667}}}$

$= \frac{270.8143}{\sqrt{10972.1533}} = 2$

$t_{0.025,10} = 2.228$

since $|t| = 2.585 > 2.228$, reject null hypo

d. when $x = 0.2$,

$y = 80.853 + 270.814(0.20)$

$= 135.016$

$t_{0.05,10} = 1.812$

$CI = y \pm t_{\frac{\alpha}{2},df} \cdot s\sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}$

$= 135.016 \pm 1.812 \times 59.8689 \times \sqrt{\frac{1}{12} + \frac{(0.20 - 0.28333)^2}{0.32667}}$

$= 135.016 \pm 35.083$

$= [99.933, 170.099]$